REVIEW ANALYSIS & INDUSTRY BENCHMARK REPORT

SupportOS+

The external code review, decoded against source-grounded evidence: three findings verified, one prescription weighed against the 2025/26 engineering standard, and a combined eleven-phase roadmap.

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Executive Summary

This report was commissioned to answer three questions that an external code review raised but did not fully answer. First, what is the reviewing developer actually telling us between the lines - what is his working model of software, and what did his review silently skip? Second, how does the SupportOS+ codebase compare, honestly and in detail, against how the strongest engineering teams build desktop applications in 2025 and 2026? Third, what should be done about all of it - in what order, at what cost, and with what explicitly ruled out? The report is grounded in two evidence streams: a direct verification pass over the repository source (file-and-line citations throughout) and twelve live web research queries executed on October 27, 2026, whose sources are listed in Appendix B.

The headline findings are symmetrical and uncomfortable for everyone involved, which is usually the sign of an honest audit. On one side, all three of the reviewer's factual observations are real: customers do re-sync in a way that looks like a full fetch on app open, there is no per-resource sync control anywhere in the UI, and the Organizations page does freeze on first open. Every one of those was confirmed against the source, and Chapter 3 documents the exact mechanisms. On the other side, the reviewer's single architectural prescription - replace raw SQL with SeaORM and reorganize models into a models folder - targets the right pain with the wrong tool, in a direction the broader Rust community moved away from for this exact class of application. And the review's loudest property is its silence: it says nothing about the project's genuinely strong engineering (727 tests, hardened CI, a serious security posture) and nothing about the areas where the project genuinely trails the 2025/26 standard (accessibility, multi-platform distribution, artifact attestation, a compile-time-checked data layer, and adversarial testing).

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factual claims from the external review verified against source code

One correction to our own prior analysis belongs in the opening chapter, because it demonstrates the standard this report holds itself to. Earlier internal discussion hypothesized that the Help Scout customers endpoint silently ignores the modifiedSince filter, which would fully explain the reviewer's symptom. Live research against the official Help Scout API documentation disproved that hypothesis: the v3 List Customers endpoint documents modifiedSince as a supported parameter, and the client code sends it correctly on incremental passes. The residual explanations for the reviewer's observation are narrower and more interesting, and they are ranked with evidence in Chapter 3. Beliefs updated on evidence is precisely the behavior this project's handover documentation should model.

1,692

raw SQL SELECT statements across the core crate - the real number behind the reviewer's ORM instinct

The recommended course of action is a merged, eleven-phase roadmap presented in Chapter 7. It absorbs everything valid in the external review (repository-layer consolidation, model separation, per-resource and single-resource sync, query performance work) and everything the review missed (accessibility activation, artifact attestations and SBOM, Windows and macOS distribution with real code signing, a verified auto-update flow, property and fuzz testing, and Model Context Protocol support in the AI Center). It also contains an explicit do-not-do list, because several plausible-looking moves - a webview rewrite, a big-bang ORM migration, CRDT-based sync - would consume months while making the product worse on the dimensions that matter to its users.

The Review, Decoded

2.1 The review, verbatim

Before analyzing the reviewer, it is worth restating exactly what he wrote, because the analysis in this chapter depends on precise wording. The review consisted of four blocks, reproduced here faithfully:

> *Follow this coding principles: DRY, don't use deep nesting, don't use arbitrary naming, prioritize readability.*
>
> *Architectural Decision: Use sea orm instead of writing sql queries everywhere. Split codes into it's own file: eg: customers model should be on customers file, all db model should be in models folder and vice versa.*
>
> *Syncing issues: Customers data is not syncing correctly. It fetches all the customers every times the app opens.*
>
> *Syncing improvements (if webhook relay is not configured): When to mark a resource synced? Fetch all the data once and mark as synced. Add options to sync resources. ex: I should be able to sync just all the customer resources and and also be able to sync a single customer resource. this options to sync needs to be on every resources. UI improvements: UI gets frozen when app loads large lists of data. ex: opening Organizations page freezes for a moment for the first time.*

2.2 Reading between the lines: the reviewer's working model

A review is a fingerprint of the reviewer's mental model, and this one carries several clear impressions. The recommendation of SeaORM specifically - rather than Diesel, sqlx, or a repository layer - is the strongest signal. SeaORM is the Rust ORM most familiar to developers arriving from Node.js ecosystems (Sequelize, TypeORM) or from Laravel's Eloquent, and it is the ORM featured in most tutorial content. Its defining appeal is exactly what the reviewer asks for next: entities in their own files, a models folder, framework-managed queries. Together with the MVC-shaped phrasing (customers model should be on customers file), this describes a developer whose formative experience is layered web applications, reviewing a local-first desktop application through a web-CRUD lens.

The second signal is the shape of the factual reports. Each is a symptom with no attached mechanism, no log excerpt, no reproduction steps, and no hypothesized root cause: it fetches all the customers every times the app opens; the UI gets frozen. A reviewer working at root-cause depth would typically attach the sync checkpoint state, the since value actually sent, the page counts observed, or a profiler stack for the freeze. The third signal is the preamble of generic principles (DRY, no deep nesting, readability) without a single file-and-line citation. That combination - broad principles, symptom reports, framework-shaped prescription - is characteristic of a competent mid-level web developer performing a skim-level pass and reporting where friction registered, not where defects live.

It is equally important to record what the review is not. It is not careless: all three observations turned out to be true, which is a better hit rate than many external reviews achieve. And it contains one genuinely excellent question - when to mark a resource synced - which goes to the heart of sync-engine semantics and which Chapter 3 answers in detail, because the project already implements a precise per-resource checkpoint policy that is simply invisible to users. The review also proposes per-resource and single-resource sync controls, which is a correct product instinct and matches the roadmap this project had already drafted.

2.3 What the review never mentions - and what that silence means

The review touches four areas: code style, data access architecture, sync behavior, and one UI performance symptom. It does not mention security (the project ships envelope encryption with AES-GCM, scrypt key derivation, restrictive file modes, secure deletion, and a Landlock/seccomp sandbox that CI actually enforces), does not mention the 727-test suite or the CI gates (clippy with warnings denied workspace-wide plus a feature-powerset lane, cargo-deny advisories, licenses and duplicate-version bans, a no-simulation gate, packaging smoke tests), and does not mention the accessibility, distribution, attestation, or AI-protocol gaps that Chapter 4 shows are where the project most visibly trails the current standard. There are two possible explanations: the reviewer did not look at those areas, or he looked and found them unremarkable. Either way, the review's coverage is a scope statement, and treating it as a complete map of the project would be the single most expensive mistake available to the team accepting it.

2.4 Claim-by-claim scorecard

Scorecard: accuracy versus depth of the external review

Review claim	Factually accurate?	Root cause identified?	Proposed fix sound?
DRY / nesting / naming issues	Yes - verified (Ch. 3.3)	No - no citations given	Partially - principles only
Use SeaORM instead of raw SQL	Pain real (1,692 SELECTs)	No - names no constraint	No - see Ch. 4.2 risk table
Models should live in models/	Yes - no models dir exists	n/a	Yes - adopted in roadmap
Customers re-fetched on every open	Symptom plausible; mechanism narrower	No - and his implied cause is disproven	Instrument first (Ch. 3.1)
Per-resource sync controls missing	Yes - only initial/incremental exist	Yes - correct observation	Yes - matches roadmap
Organizations page freezes	Yes - two stacked causes	No - no mechanism given	n/a - none proposed

Table 2.1 - The external review, graded. Accurate observations, shallow mechanisms, one wrong prescription.

Ground Truth: What the Code Actually Shows

3.1 The customers sync claim, traced end to end

The reviewer's central factual claim - customers are fully re-fetched every time the app opens - deserves the deepest treatment because it is the one where our own prior hypothesis required correction. The verified mechanics are these. At launch the background worker starts an adaptive auto-sync timer (workers.rs, lines 244-301; busy cadence 30 seconds backing off toward a configured ceiling). Each tick runs incremental_sync when state is not NEW (workers.rs, line 1015). The customers stage computes since as the resource's last_success_at checkpoint minus a ten-minute overlap (sync_engine.rs, lines 1588-1592 and 317-334, SYNC_OVERLAP_MINUTES defined at line 59), and passes it to sync_customers, which sends modifiedSince to the provider (helpscout_real.rs, lines 1863-1879). A successful stage stamps the checkpoint (record_success, sync_engine.rs, lines 277-287). Pagination follows HAL next links with absolute-URL and nextCursor fallbacks (hal_next_path, helpscout_real.rs, lines 3226-3244). The boot-time state reset to NEW fires only on an empty-to-real provider transition, and the previous provider kind is persisted across restarts (app-shell lib.rs, lines 149-180), so healthy boots do not re-trigger an initial crawl.

The decisive external fact, verified live against the official documentation on October 27, 2026, is that Help Scout's v3 List Customers endpoint documents modifiedSince (Returns only customers that were modified after this date) and cursor pagination; the v2 endpoint additionally documents sortField with a modifiedAt option. The client's outbound timestamp format (seconds precision, literal Z) matches the documented example. In short: the API-side incremental filter is real, and the client is wired to use it correctly. Our earlier hypothesis - that the server ignores the filter - is therefore withdrawn.

CORRECTED

Help Scout v3 docs do document modifiedSince for customers - the server-ignores-it hypothesis is withdrawn

That correction narrows the field of explanations for the reviewer's observation to three surviving hypotheses, all of which are consistent with the source. H1: he was testing against fresh or reset databases, where state NEW correctly triggers a full initial crawl on each first launch - exactly what the design intends, and easy to misread as every open. H2: the initial crawl on a realistically sized account never completes within its deliberate budgets - the customers stage caps channel sub-resource fetches at 400 per pass (sync_engine.rs, line 2378) and the initial crawl at 5,000 pages (line 2505) - so a resumable crawl continues across launches, which looks identical to re-fetching everyone. H3: databases migrated before the M029 checkpoint rebuild carry NULL last_success_at rows, producing one full-window incremental pass that self-heals at the next success stamp (sync_schema.rs, lines 44-66). There is also an aggravator independent of all three: upsert_customer_with rewrites every returned row unconditionally - the ON CONFLICT clause has no updated_at guard (sync.rs, lines 360-386) - so even a correctly filtered pass pays a full write, and an unfiltered one pays a full table rewrite. The fix plan is therefore instrumentation-first: log the since value, page count, and row count per customers pass; the counters will identify the surviving hypothesis in one session, and the upsert guard is worth shipping regardless.

3.2 The Organizations freeze, measured

The freeze the reviewer felt is real and has two stacked causes. The first is query cost. list_organizations returns up to 200 rows (people_store.rs, lines 6-36), and each row carries two correlated COUNT subqueries joined through the condition (cu.organization_id = o.id OR (cu.organization_id IS NULL AND cu.organization = o.name COLLATE NOCASE)). The customers table carries no index on organization_id - the column exists (mirror_parity.rs, line 31) but the only organization-related index in the codebase is idx_organizations_name - so the OR-condition forces table scans of customers and conversations per organization row. The second cause is execution context. The page calls reload() on first open (organizations.rs, lines 64-104; loaded starts false) and with_read runs the closure synchronously on the UI thread (core db.rs, lines 184-198), inside the egui frame. A cold query that costs hundreds of milliseconds is experienced as a frozen moment. Neither cause is subtle, and both have cheap fixes: an index plus an aggregate rewrite for the query, and an off-thread load with a spinner state for the frame. The egui framework virtualizes long lists through show_rows, already used on the customers and search pages, so widget volume is not the issue here.

3.3 The structural claims: DRY, nesting, naming

The reviewer's stylistic complaints are directionally correct and the numbers are large enough to be unambiguous. The core crate is 101,776 lines across roughly 120 files; raw SQL appears 1,692 times as SELECT alone; the five largest files are workers.rs at 5,235 lines, ai_pipeline.rs at 3,913, helpscout_real.rs at 3,692, outreach.rs at 3,190, and segment.rs at 3,171. Concrete duplication receipts include the identical six-line LIKE-escape block repeated in list_organizations and list_customers_summary, and the same CREATE INDEX statement for idx_organizations_name declared in three separate files (sync_schema.rs line 296, search.rs line 248, db_breadth.rs line 851). Schema DDL is scattered across at least five modules (migrations.rs, mirror_tables.rs, sync_schema.rs, search.rs, db_breadth.rs). Deep nesting reaches four to five levels inside sync_customers (loop, then if, then match, then for, then match). Opaque module names - db_breadth, mirror_parity, it_sync, it_helpscout - are fair targets for renaming. None of this is cosmetic: scattered SQL and DDL is exactly what makes entity drift invisible until a runtime test catches it, and that pain is what the reviewer was feeling when he reached for an ORM.

Structural measurements behind the style complaints

Measurement	Value	Where
Core crate total lines	101,776	crates/core/src
Raw SELECT occurrences	1,692	workspace grep
Largest file	5,235 lines (workers.rs)	crates/core/src/workers.rs
SQL-densest file	126 SELECTs (segment.rs)	crates/core/src/segment.rs
Same index DDL declared	3 times	sync_schema / search / db_breadth
models/ or repo/ folder	does not exist	crates/core/src

Table 3.1 - The numbers behind the reviewer's instinct that the data layer needs consolidation.

3.4 What the review did not flag because it is already strong

Completeness requires recording the other side of the ledger, which the review ignored entirely. The project runs 727 tests in-tree, including mirror-parity tests that assert SQL behavior against both code paths. CI on every push and pull request enforces rustfmt, clippy with warnings denied across the workspace and again with the sandbox feature enabled, the full test suite, cargo-deny across advisories, licenses, and duplicate crate versions, an icon-integrity check, and a no-simulation data gate; every GitHub Action is SHA-pinned and credentials are not persisted. The packaging lane builds the AppImage and .deb in CI and smoke-tests the boot chain without a display. The security posture - AES-GCM envelope encryption, scrypt derivation, 0600 file modes, secure deletion, HMAC webhook verification, a Landlock/seccomp sandbox exercised on real kernels - exceeds what most shipped desktop applications do. SQLite is opened with WAL, foreign keys on, synchronous NORMAL, and a five-second busy timeout. TLS is rustls. None of this excuses the gaps in Chapter 4; it does mean the baseline the reviewer was improving on was far above the one his review implies.

The Industry Standard, Researched

Everything in this chapter comes from live research performed on October 27, 2026; numbered sources appear in Appendix B. The purpose is not to name-drop tools but to establish, for each contested decision, what the current mainstream of strong engineering teams actually does - and then score SupportOS+ against it.

4.1 GUI framework choice: mainstream, but half-activated

The 2025 Rust GUI landscape has no dominant framework: surveys and community threads through early 2026 consistently split between egui and iced for tool-like applications, Tauri 2 for web-technology teams, Slint for embedded-adjacent work, and Dioxus for React-style ergonomics [S1][S2]. Against that backdrop, the project's egui choice is a mainstream, defensible position - particularly given the recorded history that both a Tauri and a Leptos variant were built and deliberately cut over to egui for single-binary distribution and memory footprint (decision D1, recorded in CI comments). What is not defensible is leaving the framework's accessibility integration switched off. egui exposes its widget tree through AccessKit behind a feature flag [S3][S4]; this project builds eframe with only the glow renderer feature, so screen readers see nothing. That omission mattered less before June 28, 2025. After that date the European Accessibility Act became enforceable, making accessibility a legal compliance dimension for digital products in the EU market [S5][S6]. Activating the flag and auditing keyboard navigation is a small, high-significance change - and it is Phase B of the roadmap.

June 28, 2025

date the European Accessibility Act became enforceable - accessibility is now a compliance category

4.2 The data layer: the community moved toward checked SQL, not heavier ORMs

The reviewer's SeaORM prescription reads differently in 2025/26 than it would have in 2021. The visible community trend among teams with query-heavy Rust workloads runs the other way: documented accounts of teams replacing ORMs with handwritten, compile-checked queries [S7][S8], engineers publishing why they switched from Diesel to sqlx [S9], and deep-dive comparisons concluding that sqlx's compile-time-verified SQL hits the sweet spot for teams who want safety without surrendering SQL control [S10]. SeaORM remains a reasonable choice for entity-driven CRUD services, but this codebase is not that: it contains a programmable segment/report DSL (126 SELECTs), FTS5 full-text corpora, sqlite-vec vec0 virtual tables, partial indexes, and checkpoint/cursor machinery - none of which map onto ORM entities. The research-supported answer for this codebase is the one already drafted independently: consolidate all SQL into a repository layer per domain, keep rusqlite, and adopt compile-time query checking where the toolchain supports it, so that schema drift becomes a build error instead of a runtime test failure. That achieves the reviewer's stated goal - SQL no longer scattered everywhere - at a fraction of the migration risk. Chapter 5 tabulates what a literal SeaORM big-bang would risk.

4.3 Distribution and updates: the standard is multi-platform, signed, and self-updating

The strongest Rust desktop teams treat release engineering as product infrastructure. cargo-dist - whose 0.30.x line remains actively maintained into 2026 [S11] - generates multi-platform release pipelines with installers and update manifests, and the ecosystem's expectations are visible in the friction reports: macOS refuses to execute unsigned binaries at all, pushing Developer ID signing and notarization to baseline [S12]. SupportOS+ currently ships Linux-only artifacts (AppImage and .deb) with SHA256SUMS, a CI smoke test, and a user-initiated update check that honestly reports a 404 for a private repository. Measured against the standard, three gaps follow: no Windows or macOS targets (which matters because most support teams live on those platforms), no artifact signing or attestations, and no download-verify-apply update flow. Relatedly, the supply-chain baseline for shipped binaries has moved to SLSA-style provenance: GitHub artifact attestations link a built artifact verifiably to its workflow run [S13], and SBOMs are increasingly requested by procurement. All of this is additive work - none of it requires touching application logic.

4.4 AI integration: MCP became the protocol of record

The Model Context Protocol, introduced by Anthropic in November 2024 [S14], was adopted during 2025 by OpenAI, Google, Microsoft, and AWS, and has since moved under the Linux Foundation's agentic AI umbrella [S15][S16]. For a product whose AI Center already implements provider management, embeddings, a sqlite-vec vector store, and an evaluation harness, adding an MCP client is the step that aligns it with where the industry converged - it would let the desktop assistant consume the growing ecosystem of MCP tool servers rather than only built-in integrations. The project's sqlite-vec dependency also warrants an honest note: the extension remains in an alpha-family release train (0.2.x alpha line as of late 2026) [S17], which the project's own decision log already records as risk row D4; the vector-store abstraction boundary should be preserved until a stable release train lands. Finally, the local-first architecture itself - SQLite as the primary store, explicit sync machinery, webhook-driven freshness - sits squarely on the trend line of the local-first movement, which produced 1.0 releases and managed platforms for sync engines throughout 2025 [S18][S19]. On architecture, the project is not behind the industry; it is early to where the industry is going.

4.5 Testing practice: the gap is method, not volume

Seven hundred twenty-seven tests is a respectable volume, but volume is the wrong metric, and the research is blunt on method: for a program whose core loop is parsing untrusted bytes (webhook payloads, HAL links, sanitized HTML, markdown extracts) and merging distributed state (checkpoints, cursors, channel preservation), the standard toolkit includes stateful property-based testing with proptest [S20] and coverage-guided fuzzing with cargo-fuzz over every input boundary. Neither exists in this repository - there are zero property tests and no fuzz targets. The elite-practice bar for this class of code is: fuzz everything that parses what an attacker can send, property-test everything that mutates state under concurrency or reordering, and keep example tests for regressions. That is roughly a week of work to scaffold and it permanently changes the defect profile of the sync and ingestion surfaces.

4.6 What elite code review actually covers

Finally, the research contextualizes the review itself. Current best-practice checklists for high-performing teams consistently include, beyond style and design: functional requirements, non-functional requirements (security, performance, accessibility), test strategy, error handling, observability, and documentation impact [S21][S22]. The external review covered style, design, and two behavioral symptoms - roughly the design-plus-style slice of a full checklist. That is not an indictment of the reviewer; it is a measurement of review depth, and it defines precisely where this report must go next: the areas a complete review would have covered, in both directions - the strengths the review ignored and the gaps it also ignored.

What the Reviewer Did Not Say

5.1 The silent checklist

The table below is the report's centerpiece. It lists twelve engineering dimensions, the verified state of this repository in each, and whether the external review said anything about it. Read as a whole, it shows a review that reported on the layer it was comfortable with and was silent on everything else - both the assets and the liabilities.

Twelve dimensions: repository reality versus review coverage

Dimension	Repository reality (verified)	Review said?
Security posture	AES-GCM envelope, scrypt, 0600, secure_delete, Landlock/seccomp sandbox	Nothing
CI hardening	clippy -D warnings x2 lanes, cargo-deny, SHA-pinned actions, smoke tests	Nothing
Test suite	727 in-tree tests incl. mirror parity	Nothing
Documentation	User handbook + developer handover book shipped	Nothing
Accessibility	AccessKit-capable framework, feature flag OFF	Nothing
i18n	No infrastructure; English strings hardcoded	Nothing
Platform coverage	Linux-only artifacts; no Windows/macOS lanes	Nothing
Artifact trust	SHA256SUMS only; no signing, SBOM, or attestations	Nothing
Auto-update	Check-only; no verify-and-apply flow	Nothing
Compile-time data layer	1,692 runtime-only SQL statements	Indirectly (via SeaORM)
Adversarial testing	0 property tests, 0 fuzz targets	Nothing
AI protocol currency	No MCP support yet; embeddings/eval present	Nothing

Table 5.1 - Review coverage versus repository reality. The review was silent on six genuine strengths and six genuine gaps.

5.2 Two truths worth keeping

Two of the reviewer's instincts deserve explicit preservation even though his prescriptions do not. First, the SQL-sprawl instinct is correct even though the ORM answer is wrong: consolidating queries into a per-domain repository layer with compile-time checking is on the roadmap and will make most of his style complaints (DRY, scattered DDL, naming) disappear as a side effect. Second, the sync-control instinct is correct and cheap: per-resource sync buttons on the Sync Health page and a single-customer sync action on the customer detail page require one public engine method (a wrapper around the private sync_resource dispatch) plus modest UI work. The project already contains the precedent - sync_single_conversation (sync_engine.rs, line 1987) - and the checkpoint semantics he asked about are already implemented and merely need to be surfaced: each resource carries its own last_success_at stamp and resumable remote cursor, written on success and rendered on the Sync Health page.

5.3 The cost of taking the prescription literally

What a literal SeaORM big-bang migration would encounter

Obstacle	Scale	Consequence if forced
Raw SQL to port	1,692 SELECTs plus writes	Months of mechanical migration
FTS5 / vec0 virtual tables	Full-text corpus + vector store	Not expressible as ORM entities; kept raw anyway
Segment report DSL	126 dynamic SELECTs	Query-builder rewrite of a programmable engine
Migration history	52 hand-written migrations on a custom _migrations table	Dual migration systems or risky consolidation
Channel-preservation logic	Audit fixes C4/M1 in sync paths	Subtle regression risk; the customers-dash bug returns
Async runtime coupling	SeaORM sits on sqlx/tokio stack	New dependency surface for marginal gain

Table 5.2 - Why the reviewer's architectural prescription is rejected while his pain points are accepted.

None of this means the reviewer's architecture sense is worthless; it means his prescription optimizes for familiarity rather than for this system's constraints. The measure of a good handover decision is that it survives contact with the constraints, and the roadmap in Chapter 7 is built to take the pain seriously while respecting what the code already learned the hard way - every audit-fix comment in the sync paths is a scar from a bug that a well-meaning rewrite would reintroduce.

The Best-Engineer Lens

6.1 Principles applied, not recited

Strip away fashion and the strongest engineers share a small set of working principles. Root cause before remedy: the reviewer reported symptoms; the roadmap starts with instrumentation and EXPLAIN QUERY PLAN before touching anything. Measure before optimize: the Organizations fix begins with a query plan, not an index guess. Non-functional requirements are requirements: the EAA made accessibility a compliance dimension, so the a11y flag is no longer optional polish. The supply chain is part of the product: SHA256SUMS answered 2020's question; attestations and SBOM answer 2026's. Developer experience is a multiplier: the repository-layer split is not cosmetics, it is what makes the next hundred changes cheaper and the next hire productive. And documentation is a deliverable: this project has already shipped a user handbook and a developer handover book, which is rarer than any framework choice in this report.

6.2 The staff-engineer scorecard

SupportOS+ against the staff-engineer bar, dimension by dimension

Dimension	Bar	Verdict
CI gates	Every push gated; warnings denied; deps audited	Meets bar
Security engineering	Threat-modeled, defense in depth, sandboxed	Exceeds bar for this class
Test strategy	Example + property + fuzz on input boundaries	Below bar (0 property, 0 fuzz)
Data layer	Compile-time-checked queries; typed IDs and states	Below bar (runtime-only checking)
Accessibility	Assistive technology works; keyboard complete	Below bar (flag off)
Distribution	All target platforms, signed, attested, updatable	Below bar (single platform, unsigned)
Observability	Structured logs, diagnostics export	Partial (fatal handler + audit log present)
Documentation	User + maintainer docs current	Meets bar (handbook + handover book)
Release engineering	Reproducible builds, smoke tests	Meets bar on Linux; partial overall
AI integration	Provider-agnostic; standard protocols	Partial (no MCP yet)

Table 6.1 - Ten dimensions scored. Four meet or exceed the bar; five are below it; one is partial.

6.3 What a staff engineer would do in week one

The answer is almost boring, and that is the point: nothing in the researched standard suggests a rewrite, a framework migration, or an architectural reset. Week one is Phase 0 plus Phase 1 of the roadmap - instrument the customers sync pass and log the since value and page counts; run EXPLAIN QUERY PLAN on the Organizations listing; add the missing index and rewrite the correlated subqueries; move the three heaviest page loads off the UI thread with visible loading states; add the updated_at guard to customer upserts. Every one of those is a day or less, every one is user-visible, and together they resolve all three factual findings of the external review with mechanisms rather than patches. Only after that does the structural work begin, because structure built on unmeasured behavior is how projects talk themselves into rewrites.

The Combined Roadmap

The roadmap below merges the earlier review-fix plan with the researched industry gaps into a single sequence. Phases 0 through 3 correspond to the original plan; phases A through H add what the research showed was missing. Sequencing follows one rule: user-visible correctness first, structure second, distribution and protocol currency third - because each phase makes the next one cheaper rather than harder.

The eleven-phase roadmap with effort, risk, and impact

Phase	Content	Effort	Risk	Impact
0	Instrumentation: sync-pass logging; EXPLAIN QUERY PLAN on org listing	0.5 d	None	Decides H1-H3
1	Hot fixes: org index + aggregate rewrite; upsert updated_at guard; off-thread page loads	1-2 d	Low	Resolves all 3 review findings
A	Hygiene: SECURITY.md, CONTRIBUTING, CODEOWNERS; SBOM + artifact attestations; ADR folder	1 d	None	Trust + onboarding
B	Accessibility: enable AccessKit feature; keyboard audit; Fluent i18n scaffolding	2-4 d	Low	EAA compliance path
C	Typed data layer: repo/ + models/ split; compile-time query checking; newtype IDs; typed sync-state enum	4-6 d	Medium	Kills SQL sprawl + drift
D	Sync options: public per-resource sync API; Sync Health buttons; single-customer sync	2-3 d	Low	Review request delivered
E	Adversarial testing: proptest on sync/merge; cargo-fuzz on webhook/HAL/sanitize	3-4 d	Low	Defect profile change
F	Platforms: Windows/macOS lanes with real code signing; Flatpak manifest	4-8 d + certs	Medium	Market coverage
G	Auto-update: download, verify, apply; signed manifests	2-3 d	Medium	Retention + safety
H	AI currency: MCP client; streaming; structured outputs	3-5 d	Low	Protocol alignment
Gate	SeaORM evaluation - only if Phase C leaves entity-drift pain	0	Held	Expect never

Table 7.1 - The combined roadmap. Certificates: Apple Developer Program and a Windows signing credential are recurring costs.

7.1 The do-not-do list

Explicitly rejected, with reasons on the record: a webview (Electron/Tauri) rewrite - the egui cutover was a deliberate, documented decision and the memory and single-binary advantages are the user's stated priority; a big-bang ORM migration - quantified in Table 5.2; CRDT or multi-device sync - the product is single-user, single-machine, and the sync engine's checkpoint design already answers the correctness question the reviewer raised; opt-in telemetry - the privacy-first posture is a feature and the update check already models honest, user-initiated behavior; and micro-optimizing build flags such as opt-level without benchmark evidence - Phase E's bench harness is the gate for that conversation. Every rejection is reversible if the facts change, and each has a named condition that would reopen it.

7.2 Success metrics

- Customers sync pass: logged since value, page count, and row count; full-window passes drop to zero after the guard ships.
- Organizations first open: p95 frame block under 16 ms after the off-thread move; query plan shows index usage.
- Accessibility: every interactive element reachable via keyboard and exposed through the AccessKit tree.
- Supply chain: every release artifact carries a SLSA-style provenance attestation and a SBOM.
- Adversarial: zero crashes from a 24-hour fuzz run across webhook, cursor, and sanitize targets.
- Platforms: signed Windows and macOS artifacts produced by the same tag pipeline that ships Linux.

Appendix A: Evidence Table

Every load-bearing claim in this report, with its primary evidence. File references are repository-relative and line numbers refer to the audited revision (main at commit bcba02c).

Evidence for load-bearing claims

Claim	Evidence
Adaptive auto-sync timer	workers.rs:244-301; cadence 30 s busy, ceiling-configured
Incremental customers stage uses since	sync_engine.rs:1588-1602; SYNC_OVERLAP_MINUTES at :59
since derives from checkpoint minus overlap	sync_engine.rs:317-334
Provider sends modifiedSince	helpscout_real.rs:1863-1879
Success stamps checkpoint	sync_engine.rs:277-287 (record_success)
HAL next robust	helpscout_real.rs:3226-3244 (absolute URL + nextCursor fallback)
Boot reset only on empty-to-real; kind persisted	sync_engine.rs:153-184; app-shell lib.rs:149-180
Channel budget and page cap	sync_engine.rs:2378 (400); :2505 (5,000 pages)
Unconditional customer upsert	sync.rs:360-386 (no updated_at guard)
v3 customers docs include modifiedSince	developer.helpscout.com list-v3 page, verified 2026-10-27
Org listing correlated subqueries	people_store.rs:6-36 (two COUNT subqueries, OR-join, NOCASE)
No index on customers.organization_id	column: mirror_parity.rs:31; only idx_organizations_name exists
Synchronous UI-thread query	organizations.rs:64-104; core db.rs:184-198
Same index DDL three times	sync_schema.rs:296; search.rs:248; db_breadth.rs:851
1,692 SELECT statements	workspace-wide count, core plus shell

Claim	Evidence
Core crate 101,776 lines	wc -l over crates/core/src
727 tests	attribute count across crates
CI gates as described	.github/workflows/ci.yml (full read)
Release profile s/LTO/strip	Cargo.toml [profile.release] lines 79-83
No AccessKit feature	Cargo.toml:61 - eframe features = [glow]

Table A.1 - Primary evidence. Research claims additionally carry numbered sources in Appendix B.

Appendix B: Research Sources

All sources accessed October 27, 2026 via live web search and page reads. Search queries and captured snippets are archived alongside this report's build artifacts.

[S1] daily.dev - A 2025 Survey of Rust GUI Libraries (April 2025).
https://daily.dev/posts/a-2025-survey-of-rust-gui-libraries-gyvdgqpur

[S2] users.rust-lang.org - Which GUI framework should I choose (stability discussion, January 2026).
https://users.rust-lang.org/t/which-gui-framwork-should-i-choose/137581/3

[S3] egui documentation - Accessibility (AccessKit integration).
https://github.com/emilk/egui/blob/main/docs/accessibility.md

[S4] lib.rs - eframe feature flags (accesskit / accesskit-unix / accesskit-winit). https://lib.rs/crates/eframe/features

[S5] Pluralsight - EU Accessibility Act: what developers need to know in 2025 (main deadline June 28, 2025).
https://www.pluralsight.com/resources/blog/software-development/eu-accessiblity-act-2025

[S6] Corpsoft - European Accessibility Act requirements guide (enforceable June 28, 2025).
https://corpsoft.io/2026/04/15/eaa-website-requirements-guide

[S7] Medium (TheOpinionatedDev) - We replaced ORMs with handwritten Rust queries. https://medium.com/@theopinionatedev/we-replaced-orms-with-handwritten-rust-queries-the-results-were-shocking-eef821f65e78

[S8] Reddit r/rust - My take on databases with Rust (sea-orm vs diesel).
https://www.reddit.com/r/rust/comments/1e8ld5d/

[S9] LinkedIn (Nathan Reller) - Why I switched from Diesel to sqlx.
https://www.linkedin.com/posts/nathanreller_rust-database-sql-activity-7409312433338535936-JbXX

[S10] Leapcell - Diesel and SQLx: a deep dive into Rust ORMs (October 2025).
https://leapcell.io/blog/diesel-and-sqlx-a-deep-dive-into-rust-orms

[S11] crates.io - cargo-dist 0.30.4 (Shippable application packaging for Rust, release entry February 2026).
https://crates.io/crates/cargo-dist/0.30.4

[S12] users.rust-lang.org - macOS refuses unsigned executables; bundle signing required (August 2025).
https://users.rust-lang.org/t/how-to-create-macos-bundles-for-apple-silicon-they-always-seem-broken/132678/2

[S13] GitHub Community - SLSA attestations linking artifacts to Actions runs.
https://github.com/orgs/community/discussions/138249 (with codelit.io supply-chain overview

https://codelit.io/blog/supply-chain-security)

[S14] Anthropic - Introducing the Model Context Protocol (November 25, 2024). https://www.anthropic.com/news/model-context-protocol

[S15] Digital Applied - MCP adoption statistics: OpenAI, Google, Microsoft support; Linux Foundation governance. https://www.digitalapplied.com/blog/mcp-adoption-statistics-2026-model-context-protocol

[S16] Gupta, Deepak - MCP enterprise adoption guide (December 2025). https://guptadeepak.com/the-complete-guide-to-model-context-protocol-mcp-enterprise-adoption-market-trends-and-implementation-strategies

[S17] Reddit r/vectordatabase - sqlite-vec 0.2.3-alpha release notes (community fork). https://www.reddit.com/r/vectordatabase/comments/1py8905/

[S18] Electric - Electric Cloud public beta; sync engine 1.0 (April 2025). https://electric.ax/blog/2025/04/07/electric-cloud-public-beta-release

[S19] Reddit r/localfirst - sync engine ecosystem discussion (PowerSync, LiveStore, Electric). https://www.reddit.com/r/localfirst/comments/1t673z2/sync_engine_research

[S20] Readyset - Stateful property testing in Rust (proptest state machine pattern). https://readyset.io/blog/stateful-property-testing-in-rust

[S21] DevTeam.space - Code review checklist: feature, non-functional, design, formatting, documentation. https://www.devteam.space/blog/code-review-checklist

[S22] Group107 - Code review best practices 2025: security, automation, collaboration. https://group107.com/blog/code-review-best-practices